

ARCHITECTURAL OVERVIEW

metaLncRNA

A Sustainable, Mamba-Orchestrated
Ensemble Predictor for Long
Non-coding RNA Identification

Python 3.10+

Bioconda

Ollama AI

HPC-Ready

The Dual Challenge in High-Throughput Transcriptomics



Methodological Instability

Algorithmic Bias

Individual tools over-rely on specific transcript features (e.g., k-mer frequency vs. ORF length), leading to extreme discordance rates on the same dataset.

Abandoned Solutions

Prior ensemble attempts (e.g., ezLncPred) established the need for consensus but failed to maintain active development.



Software Atrophy & The Reproducibility Gap

Dependency Hell

Critical legacy tools (CNCI, CPC2, CPPred) are trapped in deprecated Python 2.7 environments.

System Failure

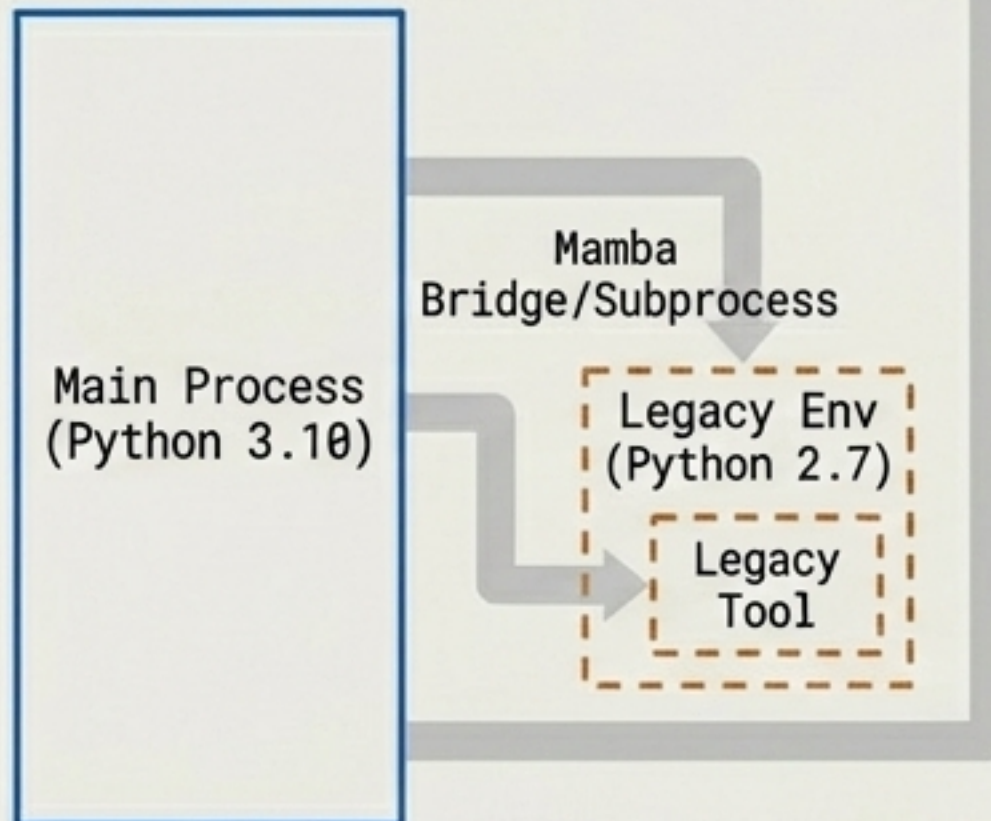
Modern 3.x frameworks cannot natively integrate these legacy tools, causing widespread failure of scientific reproducibility over time.

A High-Performance, Modular Python Framework

Environment Orchestration

Automated, transparent Mamba isolation.

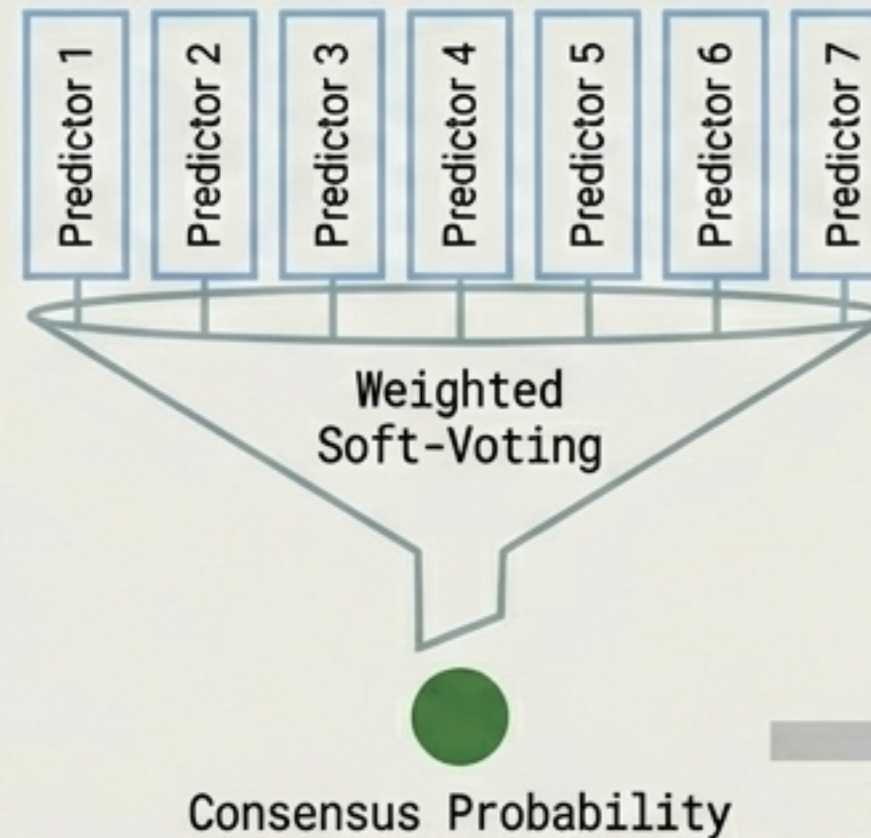
Bridges modern Python 3.10 main processes with legacy 2.7 dependencies.



The Ensemble Engine

Simultaneous, parallel execution of 7 high-performance predictors via `concurrent.futures`.

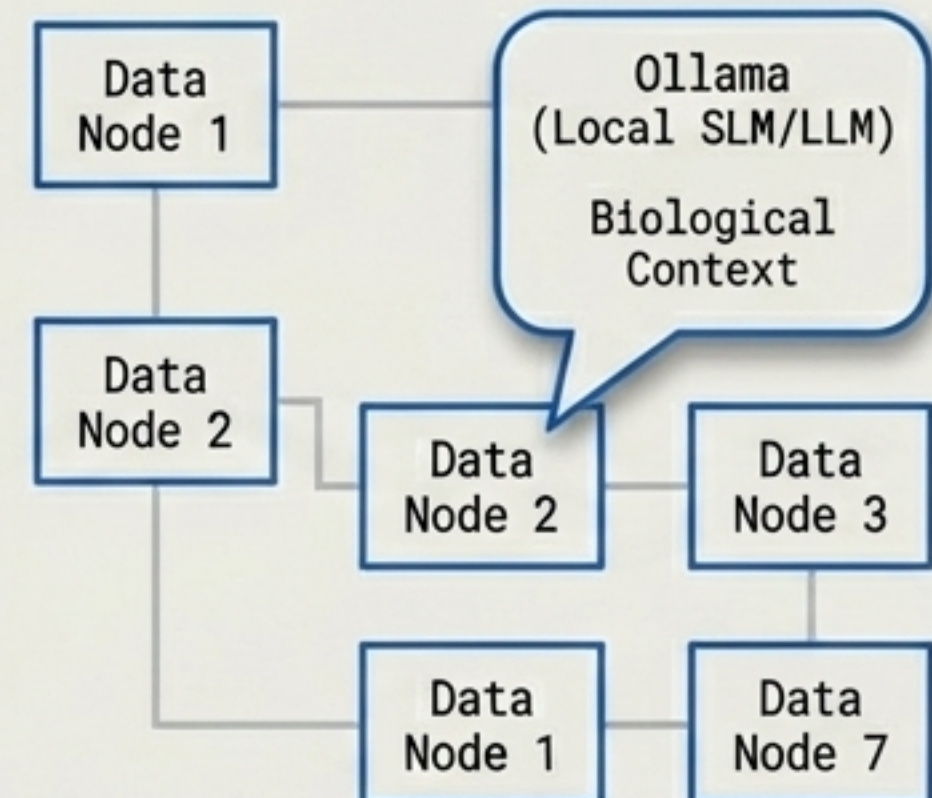
Weighted soft-voting algorithm to generate a robust consensus probability.



AI-Driven Interpretation

Local SLM/LLM integration (Ollama).

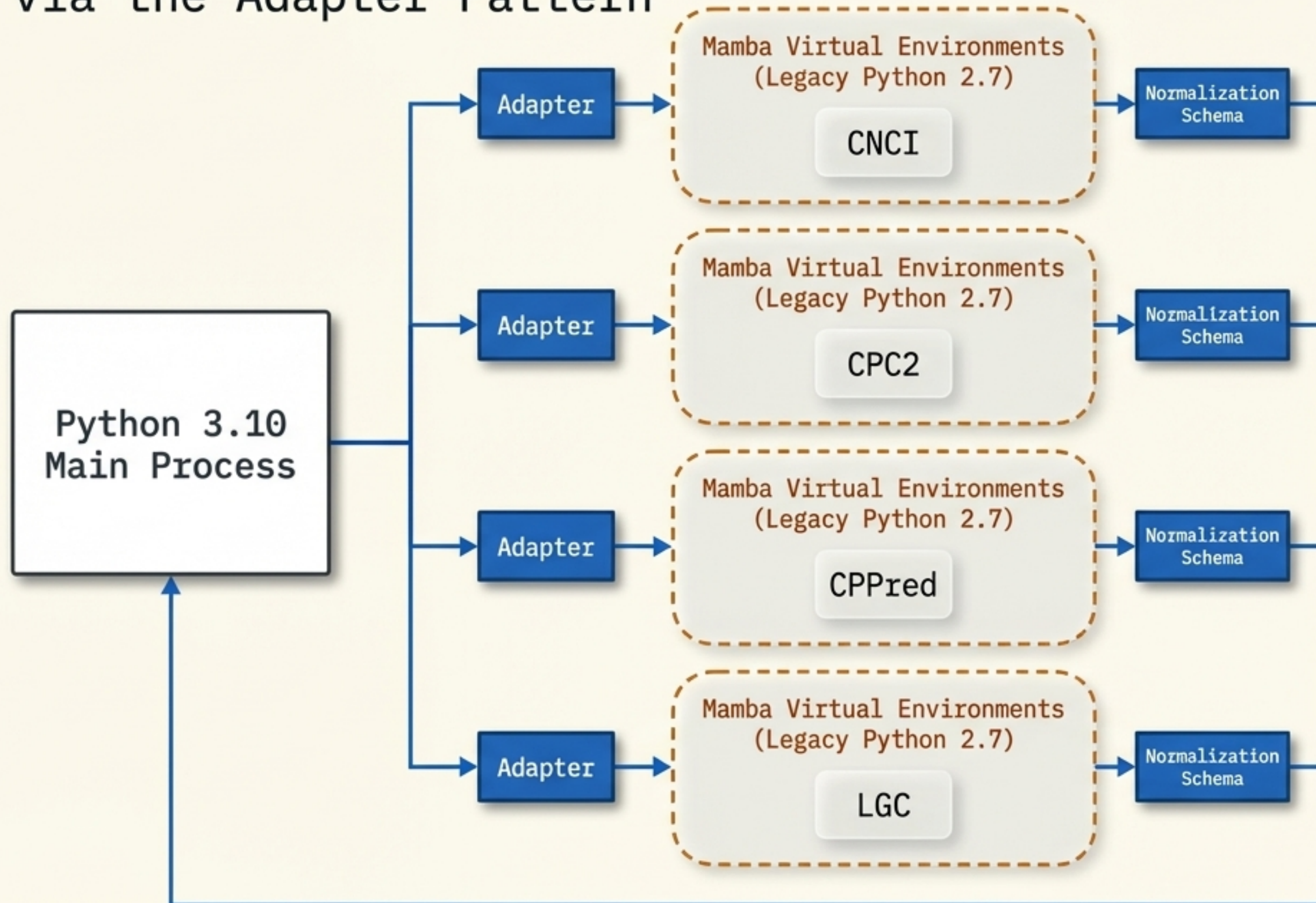
Derives natural language biological context from raw prediction conflicts without data leakage.



Covering the Computational Pillars of lncRNA Detection

Category	Integrated Tools	Underlying Methodology
Statistical Codon Bias	CPAT LGC	Logistic regression & codon bias statistics.
Machine Learning	CPC2 PLEK CNCI	Support Vector Machines (SVM) on k-mers & ORF metrics.
Integrative Features	CPPred	SVM on integrated, multi-source sequence features.
Deep Learning	RNAseam	Convolutional Neural Networks (CNN) for sequence pattern recognition.

Transparent Environment Orchestration via the Adapter Pattern



Legacy Preservation

Source code natively bundled in a third_party directory-immune to external repository deletion.

The Adapter Pattern

Wraps each tool in a dedicated execution adapter, mapping absolute paths and enforcing safe CWDs.

Normalization

Heterogeneous outputs are standardized into a unified schema before returning to the main process.

The Consensus Engine: Weighted Soft-Voting Algorithm

$$S = \frac{\sum_{i=1}^n w_i \cdot P_i}{\sum_{i=1}^n w_i}$$

Final Consensus Meta-Score

Benchmarked Coding Probability (from individual P -scores)

Benchmarked Reliability Weight (calibrated to maximize F1-scores)

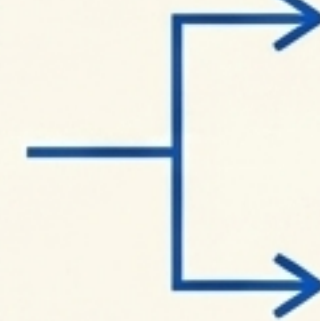
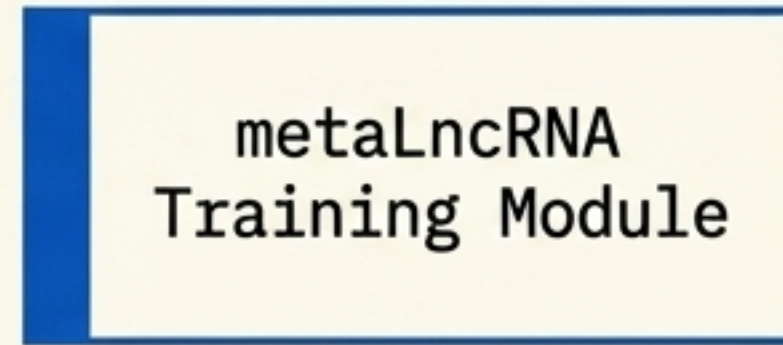
Predicted Coding Probability from individual tool i

Transcript classified as lncRNA if $S \leq 0.5$. Downcasted memory optimization ensures scalability for massive datasets.

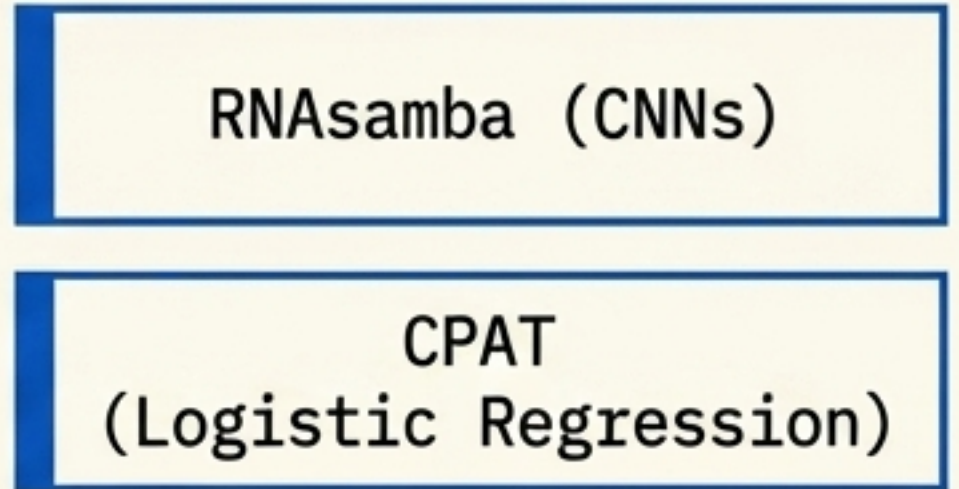
A Customizable Research Platform for Non-Model Organisms

Generating species-specific predictive models directly within the framework.

Custom FASTA Input
(Coding + Non-Coding)



Species-Specific Models



Convergence Requirements:

Input FASTA must contain at least 100 sequences and valid ORF structures for algorithm convergence.

```
$ metaLncrna train --species my_organism \  
  --cpat_coding coding.fa \  
  --cpat_noncoding noncoding.fa
```


AI-Driven Biological Insights with Zero Data Leakage

Ollama LLM Engine

- **100% Local Inference:** No cloud APIs. Proprietary transcriptomic data remains strictly on-premises.
- **Context-Aware:** Agent ingests conflicting tool probabilities to explain consensus decisions.

Standard Tier

llama3.2 (Requires 4GB RAM)

Specialist Tier

saama/openbiollm-llama3-8b (Fine-tuned on PubMed; Requires 8GB+ RAM)

User > Why did CPAT disagree with the consensus for transcript_042?

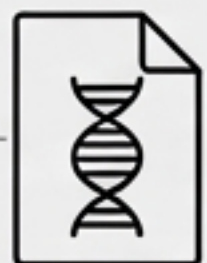
Agent > transcript_042 shows atypical codon bias triggering CPAT's coding threshold, but RNAssamba's deep learning and CPC2's k-mer analysis strongly identify non-coding structural motifs, pushing the final soft-voting meta-score to 0.32.

Standardized Outputs & Checkpoint-Aware Fault Tolerance



metaincrna_results.tsv

Unified consensus table containing individual probabilities and the meta-score (S).



predicted_lncrnas.fasta

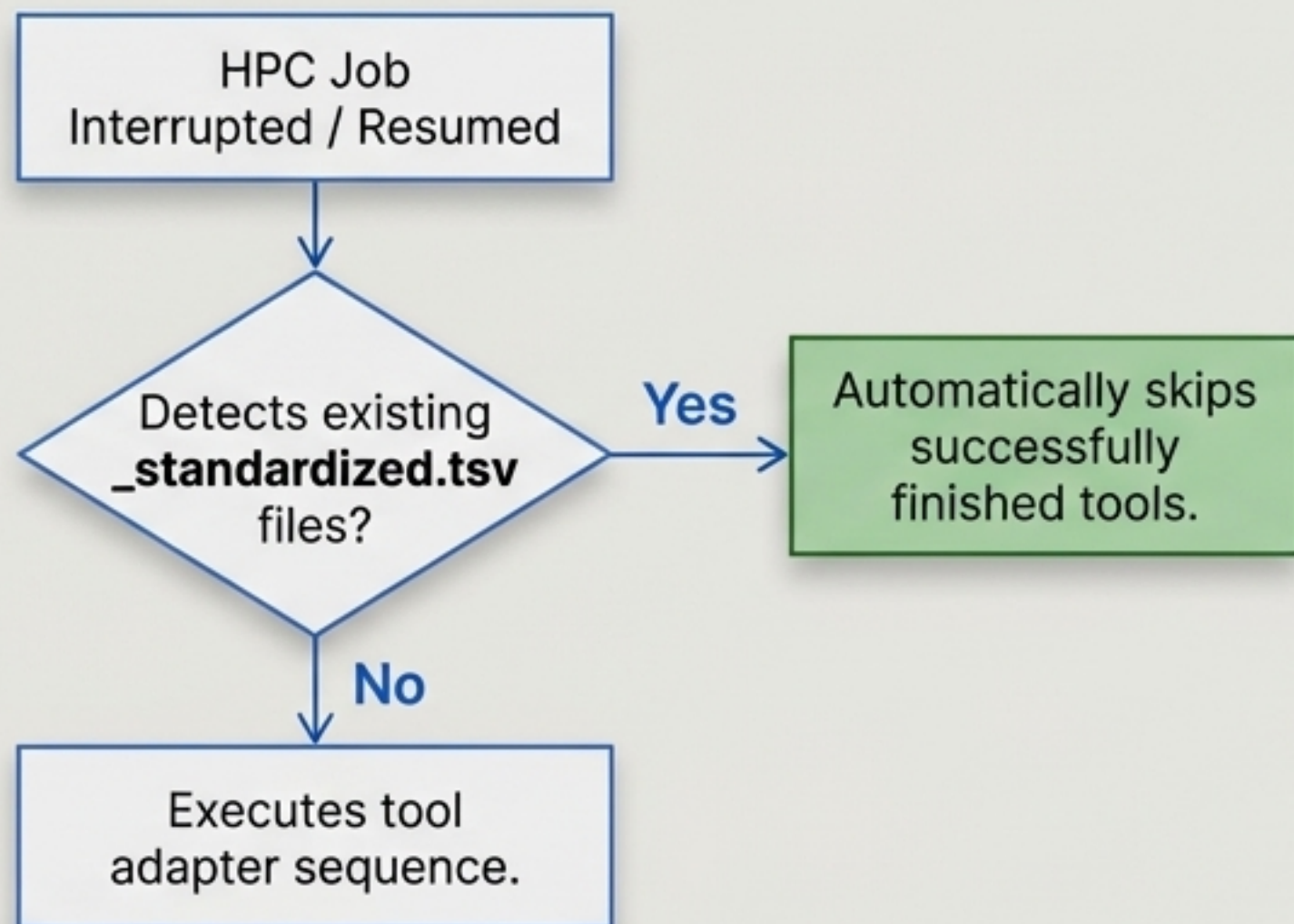
Filtered sequence file ready for downstream functional genomic analysis.



metaincrna.log

Comprehensive execution log ensuring full auditability.

Smart Checkpointing Flow



Saves critical compute hours by never repeating completed work.

Built for Omnichannel HPC & Cloud Deployment



PyPI & Bioconda

Native ecosystem installation for standard bioinformatics environments.



Pixi

Declarative, reproducible development environment via `pixi.toml`.



Docker & Singularity

Pre-configured container images for absolute reproducibility and isolation on shared HPC clusters.



Galaxy

Validated ToolShed wrapper for web-based workflows tailored to experimental biologists.

Modular Execution Workflow

1. Prepare Mamba environments & download models

```
$ metaIncrna setup
```

2. Run the 7-tool ensemble on transcript dataset

```
$ metaIncrna predict -i transcripts.fa -o output_dir
```

3. Aggregate manually (if tools run individually on distributed nodes)

```
$ metaIncrna aggregate --dir tool_outputs/
```

4. Launch interactive local AI session on consensus results

```
$ metaIncrna chat -r metaIncrna_results.tsv
```


HPC Best Practices & Operational Troubleshooting



Error: Permission Denied during container execution.

Fix: Ensure correct volume binding.

```
singularity run --bind /my/data:/data  
metaLncRNA.sif ...
```



Error: Mamba command not found.

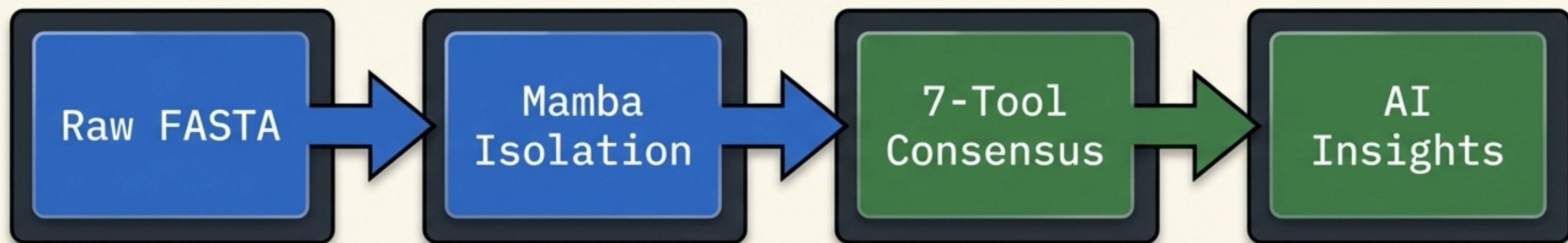
Fix: The environment orchestrator relies on Mambaforge/Conda. Ensure it is actively exported in your \$PATH.



Issue: Slow generation or crashing during AI Chat.

Fix: Specialist 8B models require significant memory. Ensure the compute node allocating the chat process has >8GB free RAM.

The Definitive Platform: From Raw Reads to Interpreted Consensus



Sustainable

Immune to software atrophy via native code bundling and automated Mamba isolation.

Rigorous

Mitigates distinct algorithmic biases via benchmarked weighted soft-voting.

Extensible

Built-in retraining modules for novel and non-model organisms.

Intelligent

The first pipeline to offer localized, zero-leakage LLM interpretation for genomic discordance.

```
$ pip install metalncrna | $ pip install 'metalncrna[agent]'
```